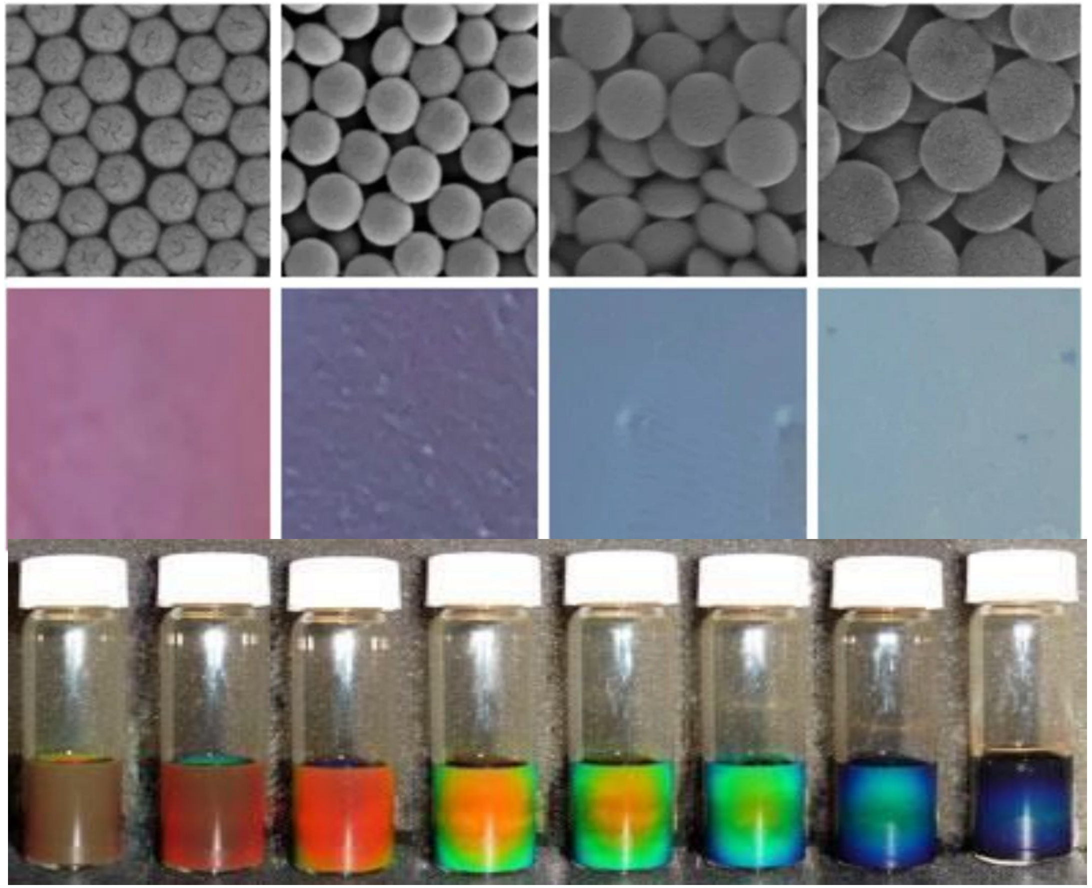
Investigation of self-assembly nanoparticles to create structural color at a water-in-oil interface

By Alexander Contreras

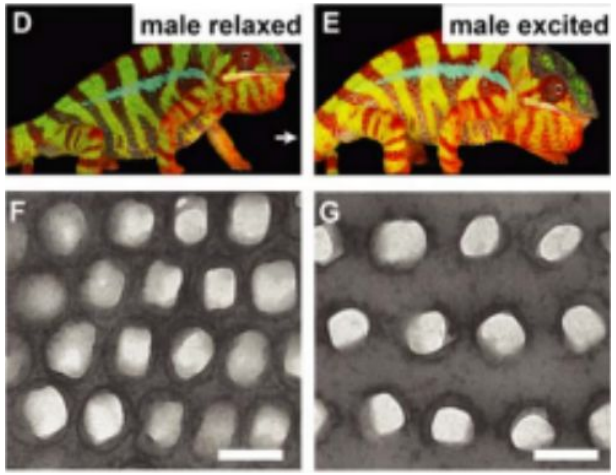


Overview

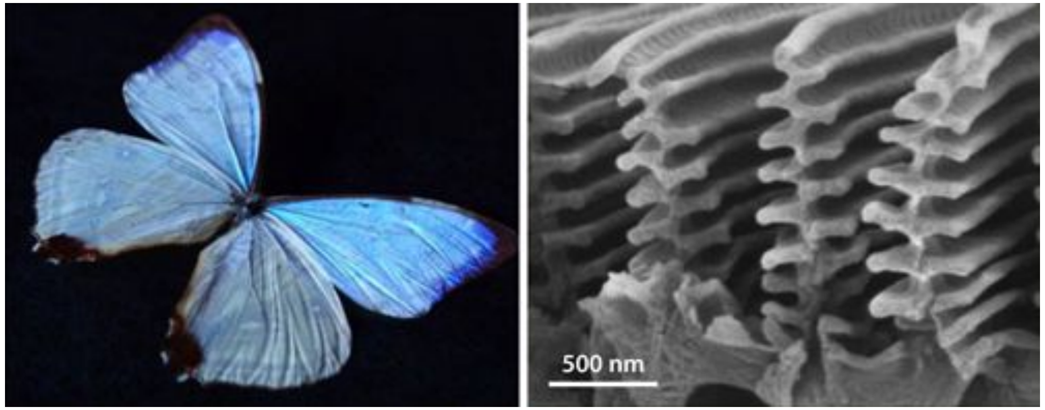
- Background Information
- Previous Studies
- Goal
- Methods
- Results
- Applications
- Conclusions
- Acknowledgements

Background Information - Pigment/Dye vs. Structural Color

Interparticle Spacing Chameleon



McDougal, Anthony, et al. "IOPscience." *Journal of Optics*, IOP Publishing, 29 May 2018, https://iopscience.iop.org/article/10.1088/2040-8986/aaff39.



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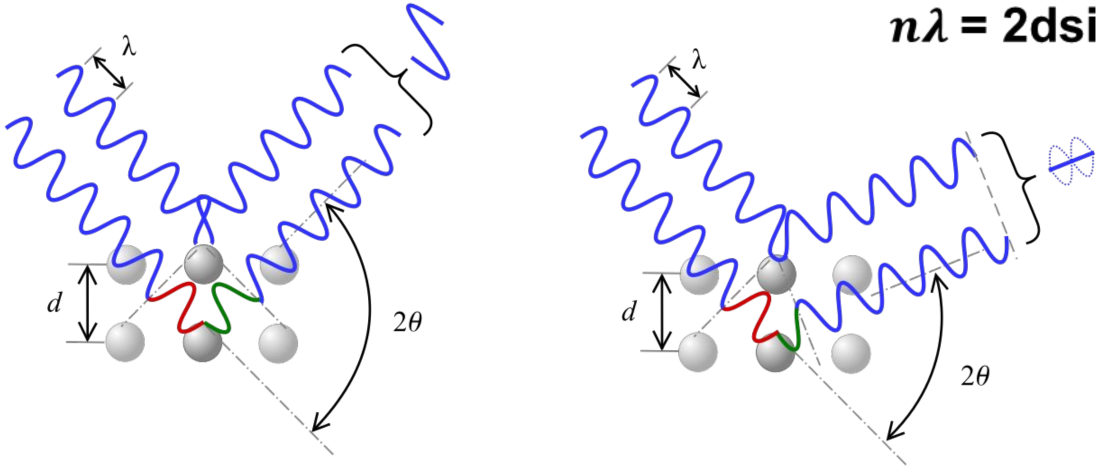
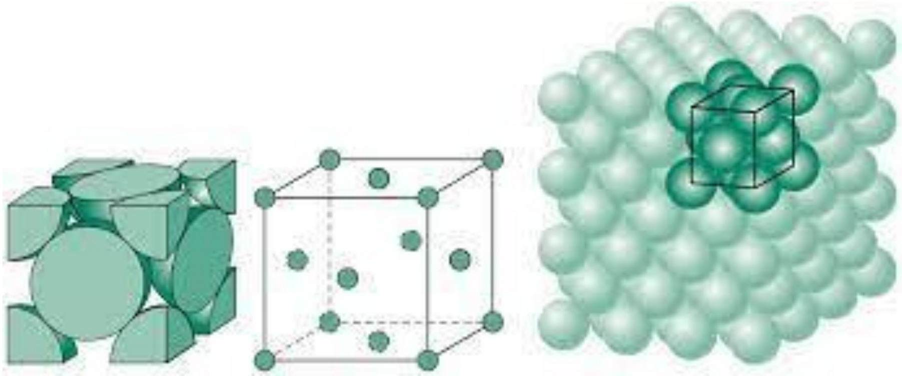
Bragg's Law – light scatters due to the successive layers of a crystal lattice by constructively interfering with wavelengths of the same frequency.

$n\lambda = 2d\sin\theta$



FCC – Face centered cubic, crystal lattice. Specific orientation and spacing of atoms.



Colloids and Surfactants

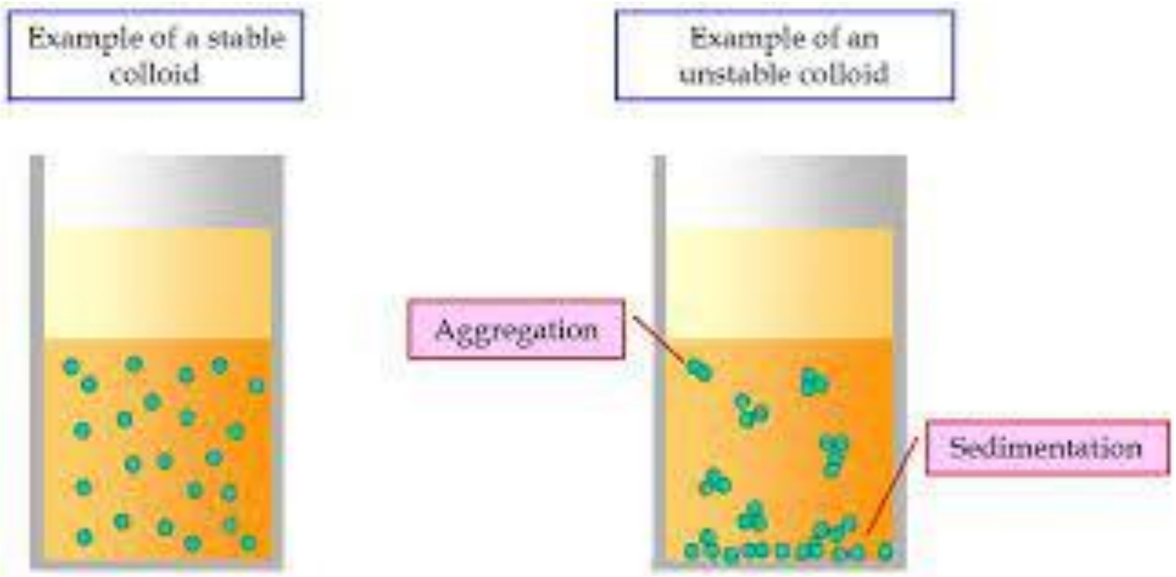
Colloids - microscopic insoluble particles suspended throughout another substances

Ex. Milk – Butterfat suspended in water
Fog – water droplets suspended in air

Surfactant - compounds that lower surface tensions between two phases, emulsifier!

Ex. Soap, detergent
Used to stabilize interfaces as well! – Blowing bubbles!

An emulsion is a colloidal mixture of two or more liquids
That normally don't mix (insoluble, miscible)



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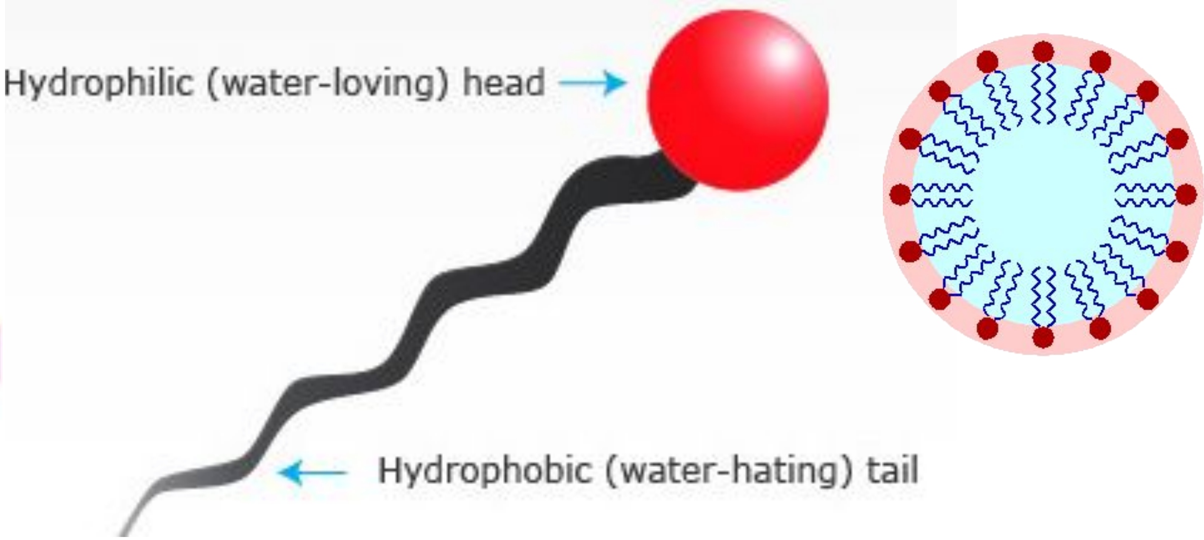

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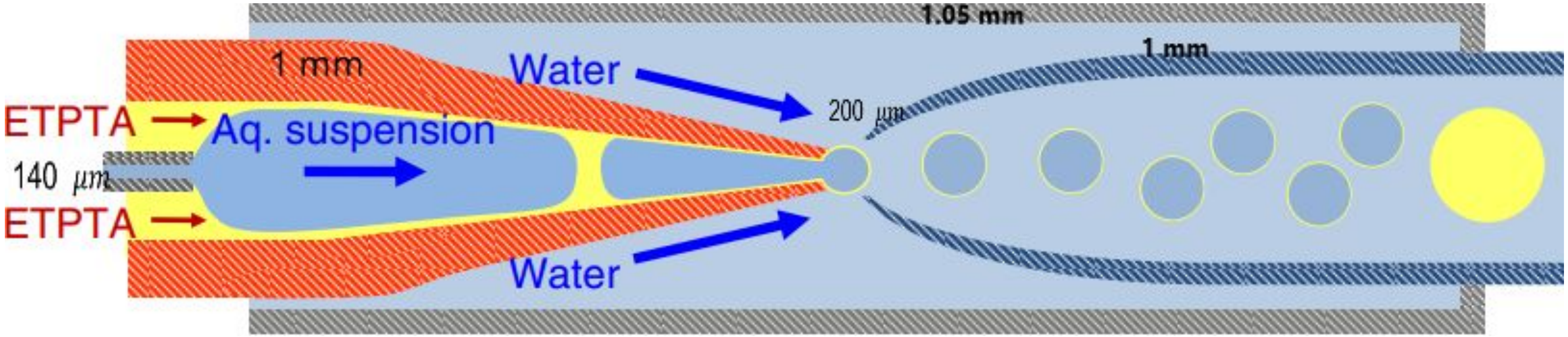
Previous Work(cont.)

Experimental Design Requirements – confine and manipulate interparticle spacing.

- Water-in-oil interfaces, easiest and most predictable environment method to modulate parameters

Double Emulsion Drops

- Aqueous suspension of Polystyrene (PS) particles with Poly(N-isopropylacrylamide) (pNIPAAm) grafted on the surfaces of -> hydrophilic
- Encapsulated in Trimethylolpropane Ethoxylate Triacrylate (ETPTA) (oil) shell.



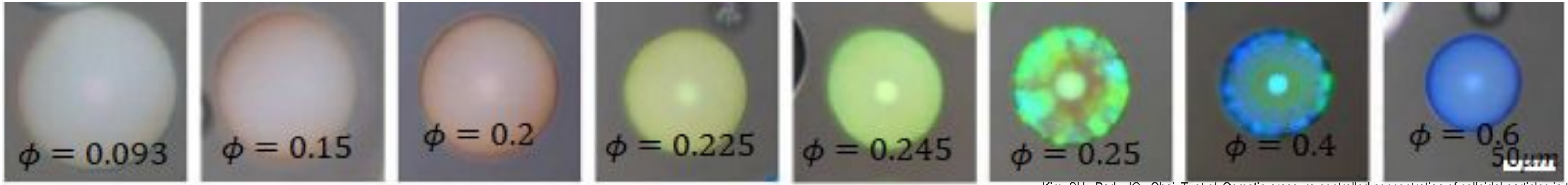
Kim, SH., Park, JG., Choi, T. *et al.* Osmotic-pressure-controlled concentration of colloidal particles in thin-shelled capsules. *Nat Commun* **5,** 3068 (2014). https://doi.org/10.1038/ncomms4068

Previous Work

$n\lambda = 2d\sin\theta$

Modulate particle concentration via osmotic pressure

- concertation in the inner phase > continuous phase = drives water out of inner phase



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Crystalline order of particles - narrowing of reflectance peaks (FWHM) with increasing ϕ. Repulsive interaction between particles due to the steric effect of the grafted pNIPAAm chains.

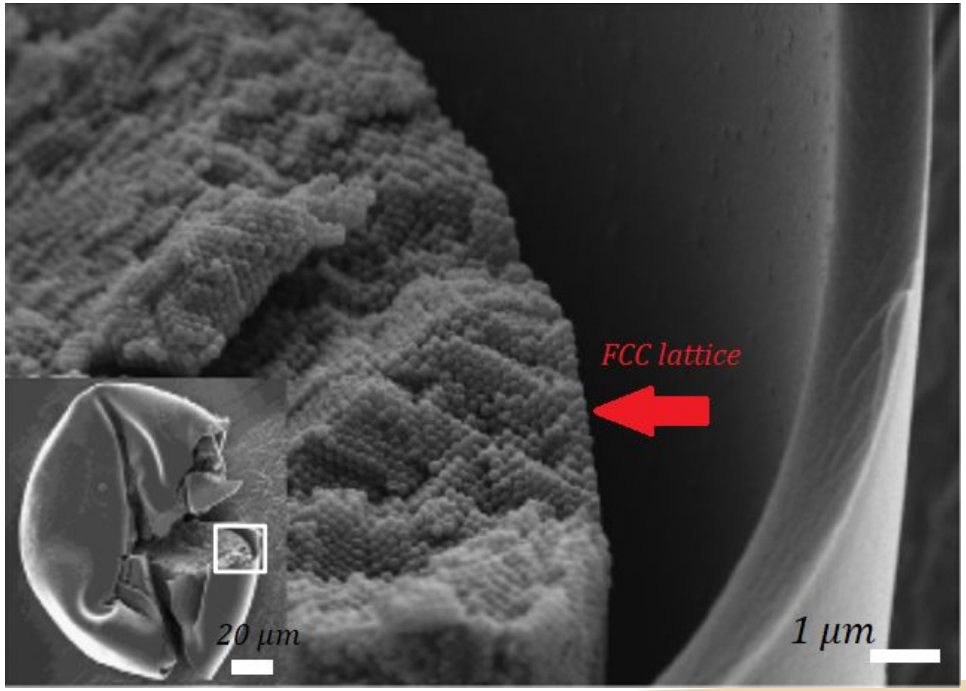
SEM scanning

- color patterns due to the FCC lattice along surface of droplet

Relationship between the color and parameters of double emulsion drop

- d is the diameter of the particle spheres
- ϕ is the volume fraction of the inner aqueous drop
- n_p and n_w is refractive index of particles and water

$$\lambda = \left(\frac{\pi}{3\sqrt{2}\phi}\right)^{1/3}\left(\frac{8}{3}\right)^{1/2} d\left(n_p{}^2\phi + n_w{}^2(1-\phi)\right)^{1/3}$$



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Goal

The goal of my project is to be able to manipulate the volume fraction of aqueous droplets of colloidal particles-in-oil to create structural color

Plan Overview

- Synthesize ~ 150 – 250nm sized particles
- Create ~ 150 - 200μm emulsion droplets
- Initial volume fraction core particles ϕ ~ 0.093
- Extremely important that they are monodispersed, (consistency in size of particles)

150 nm particles
$\phi = 0.3$
Blue shift ~ 417 nm

vs

250 nm particles
$\phi = 0.3$
Red shift ~ 695 nm

$$\lambda = \left(\frac{\pi}{3\sqrt{2}\phi}\right)^{1/3} \left(\frac{8}{3}\right)^{1/2} d\left(n_p{}^2\phi + n_w{}^2(1-\phi)\right)^{1/3}$$

Original Approach: Test whether an aqueous droplet could shrink without an oil membrane, driven only by oil-water solubility.

Research Focus: Identify oils with suitable solubility to promote outward water flux without destabilizing the droplet.

Optical Instruments:
Inverted brightfield microscope (Zeiss Axio Observer 7)
4X Objective Lens

Procedure: Place droplets in an oil reservoir and captured images every 30 minutes to track changes in droplet size.

Methods

Synthesis of TPM suspended in aqueous solution

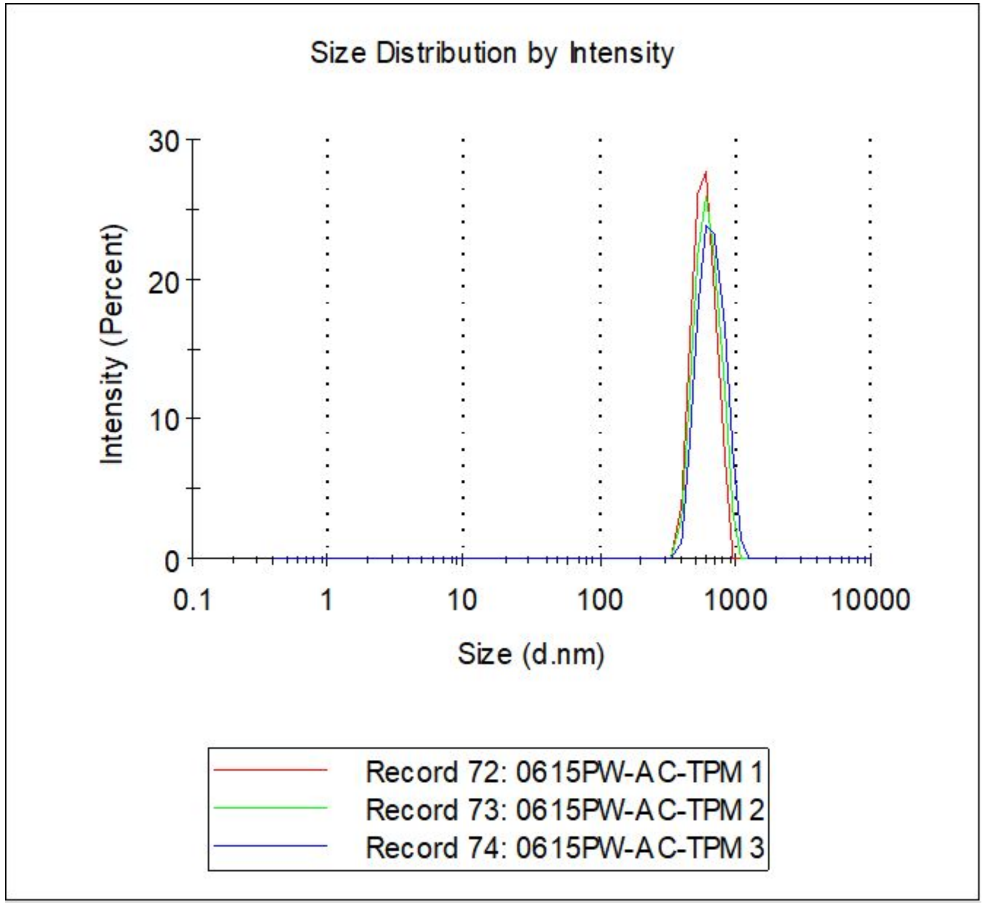
Vial 1: 9.8mL H2O + 10 μL NH4OH

Vial 2: 980 μL TPM +20 μL PI

- Sonicate both
- 27.8 μL Vial 2 to Vial 1
- ADD Surfactant
- Stir with magnetic stir for 1.5 hours,
- Under UV light for 10 minutes take out every 3 mins to shake

Dynamic light scattering to confirm

- Size and dispersity of particles



Calculate Volume Fraction

- Weigh an empty test tube
- Pipette $200\mu L$ of aq. solution
- Dry water out in oven
- Difference in weight = weight of particles
- V = D/m
- $\phi = {}^{V_{TPM}}/_{V_i}$

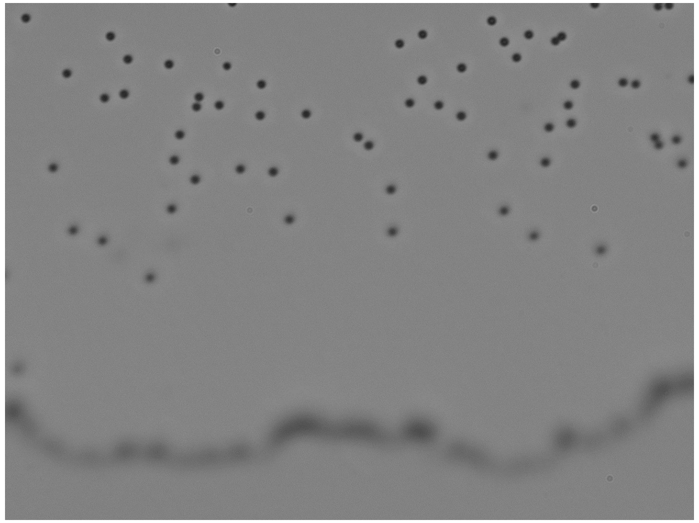
First Trials with Cyclopentane

No noticeable size change

Trials:

- Toluene – 526 mg/L
- Chloroform - 7950 mg/L

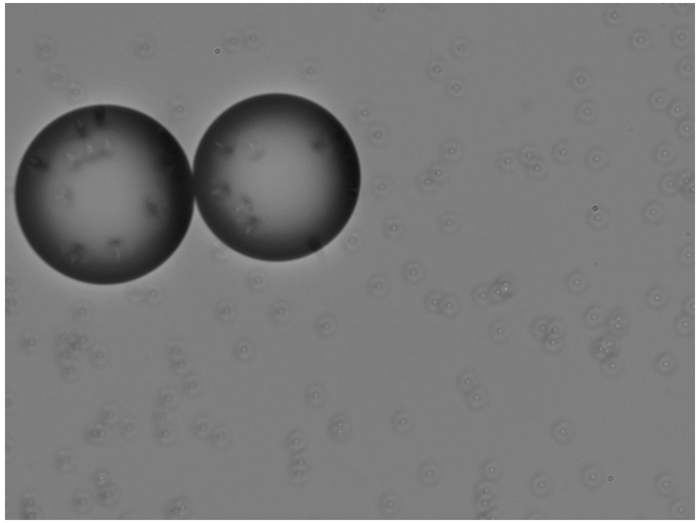
- Tried Cyclopentane – 156 mg/L



1 µm PS Droplet in Cyclopentane

- Initial ϕ ~ 0.019
- Not enough surfactant
- Unstable Interface

Inverted brightfield microscope
4X Objective Lens



1 µm PS Droplet with more CTAB Surfactant

- Stabilized interface
- No observed shrinkage
- No Coalescing between droplets

Transport rate too high?

Cyclopentane and Toluene 1:1 ratio

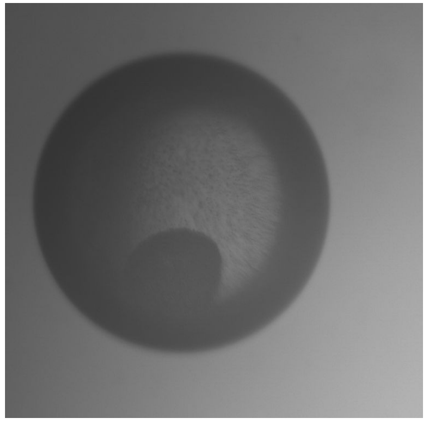
- MFM Microscope
- 4X Lens Objective
- Total volume of oil 6mL
- 100 nm PS + CTAB (Aqueous solution)
- Initial $\phi \sim 0.03$
- ($\phi \sim 0.09$ blue shift)

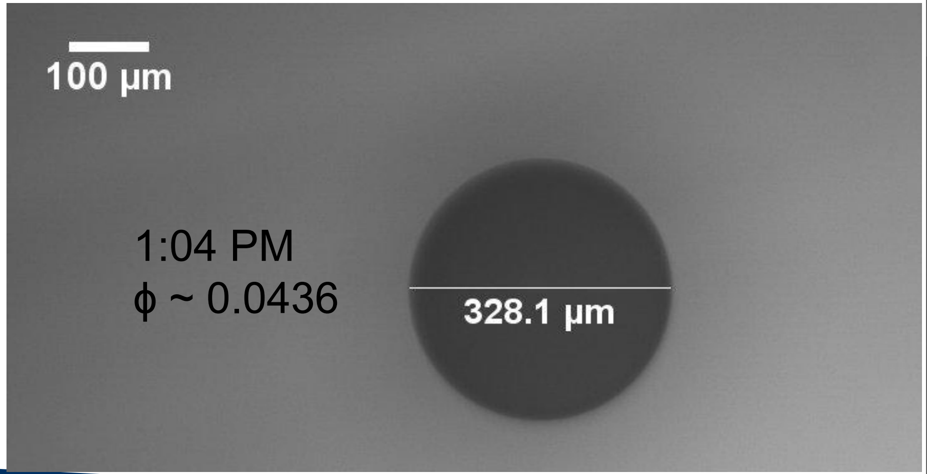
$$\phi = \phi_o\left(\frac{R_o}{R}\right)^3$$

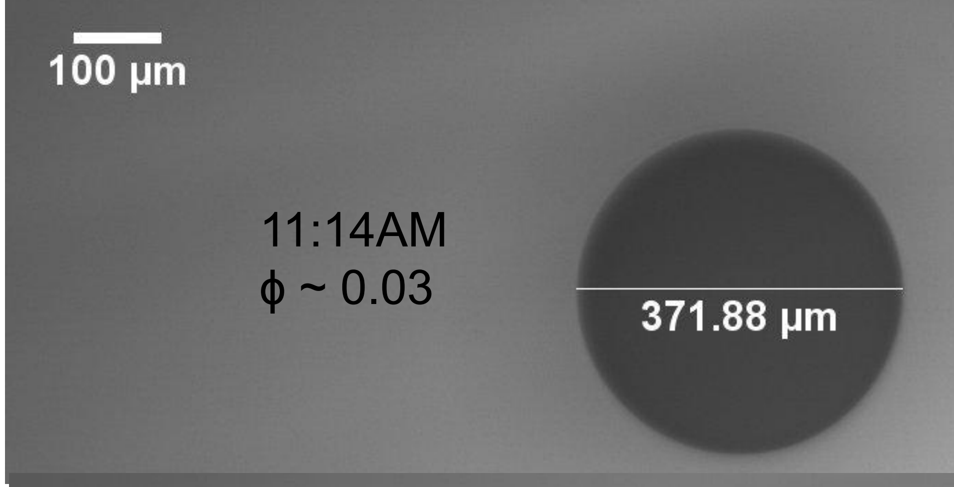
Check in every ~ 30 min add 200 μL Cyclopentane and stir with pipette

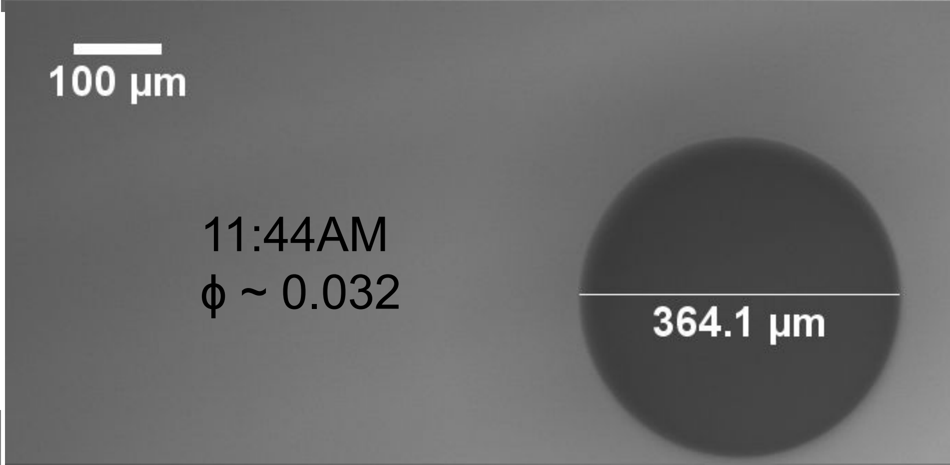
- 4th round of Cyclopentane, the droplet burst

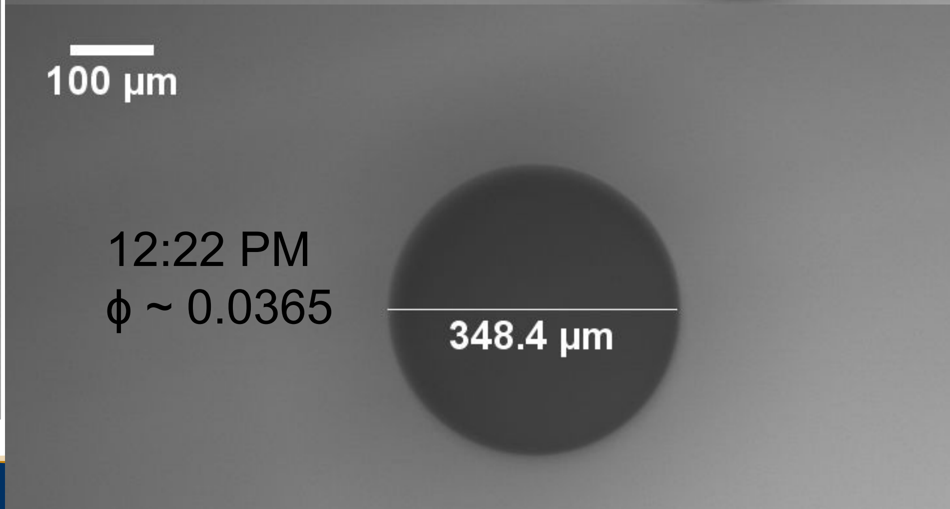
Cheerio Effect



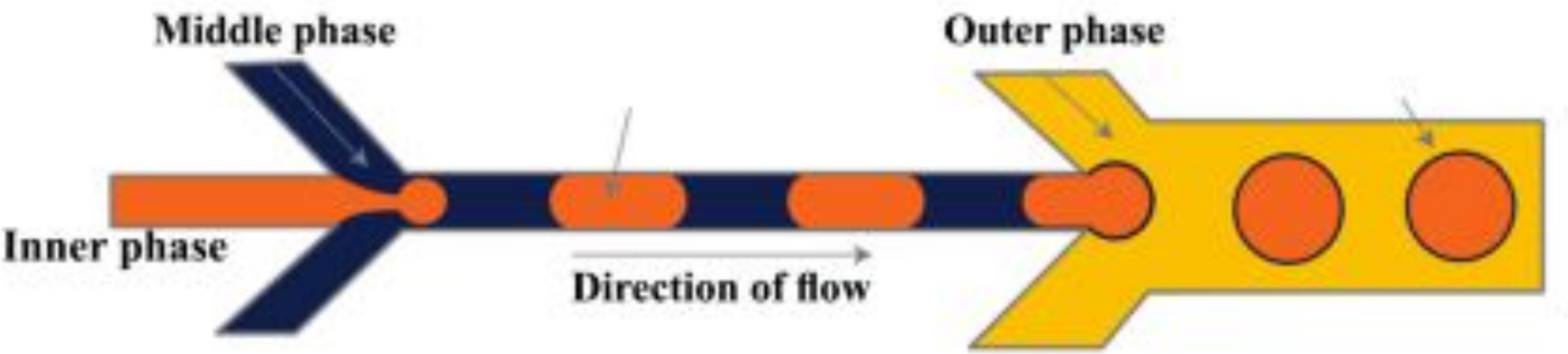








Double Emulsion Microfluidics



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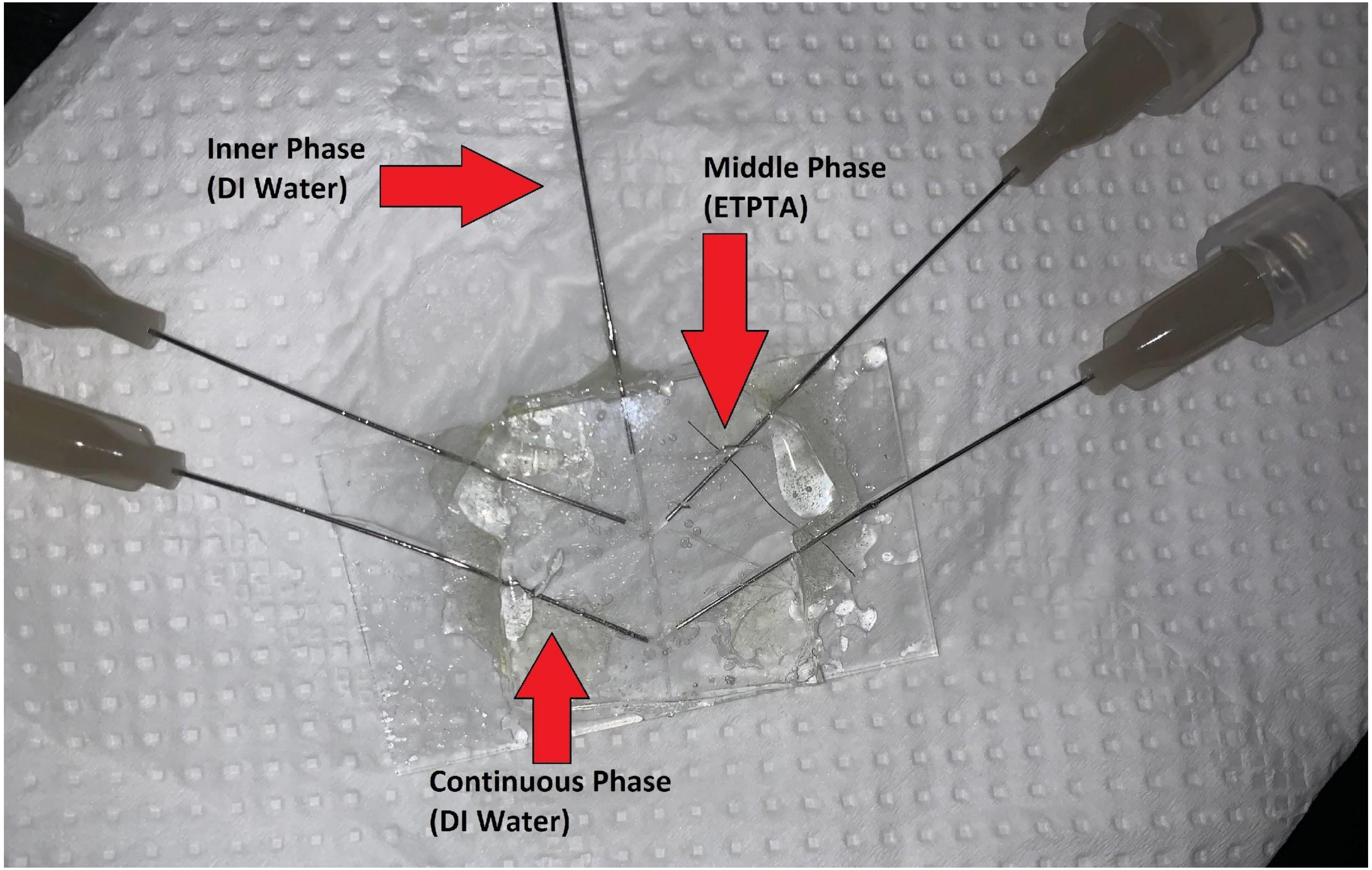
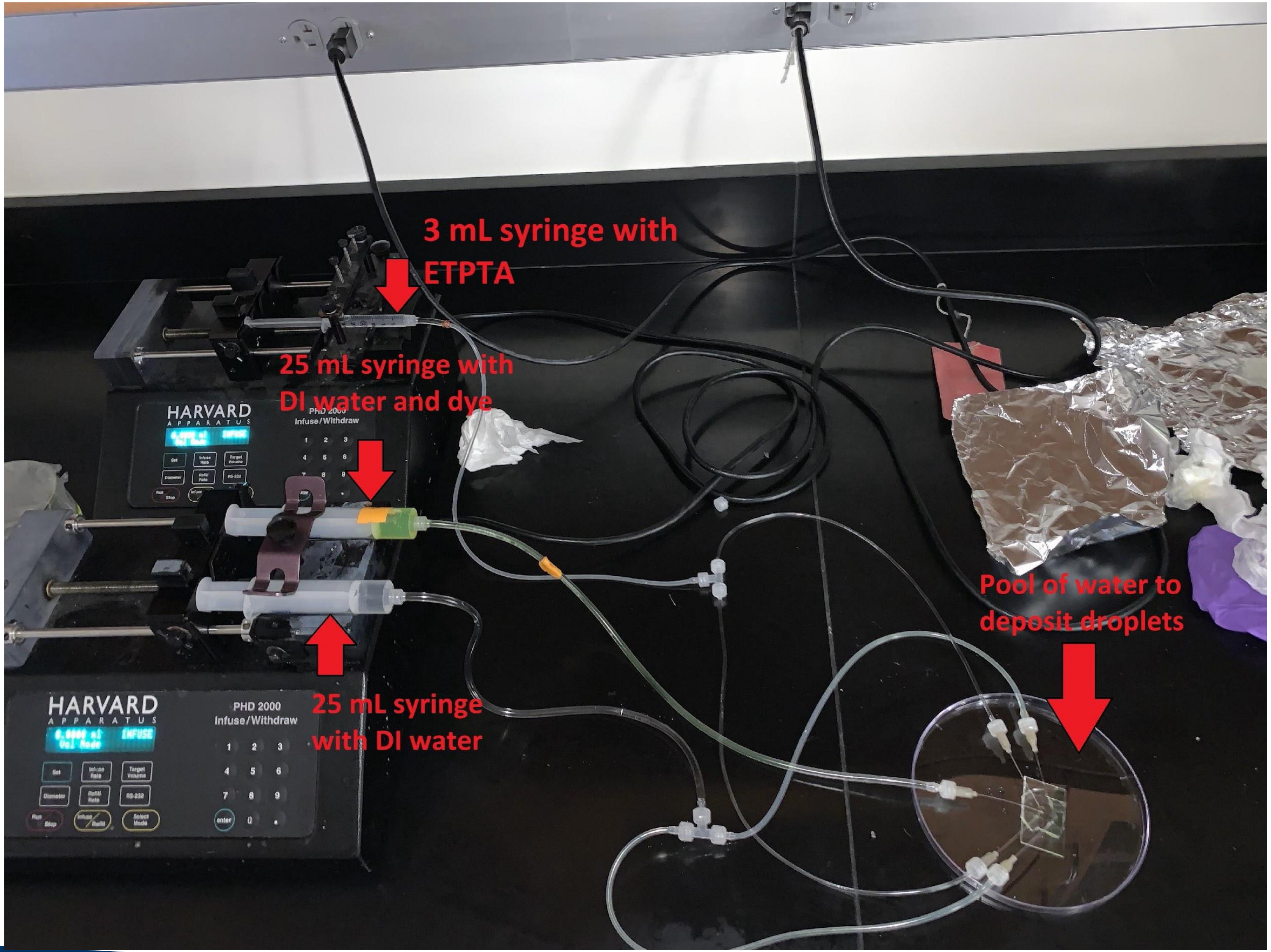
Double Emulsions PDMS



- PDMS molds using copper wire segments
- Double PDMS and Epoxy to limit leakage
- Minimal if any leakage!

Double Emulsion Setup



Inner phase rate: $250\mu L/min$

Middle phase rate: $25\mu L/\mathrm{min}$

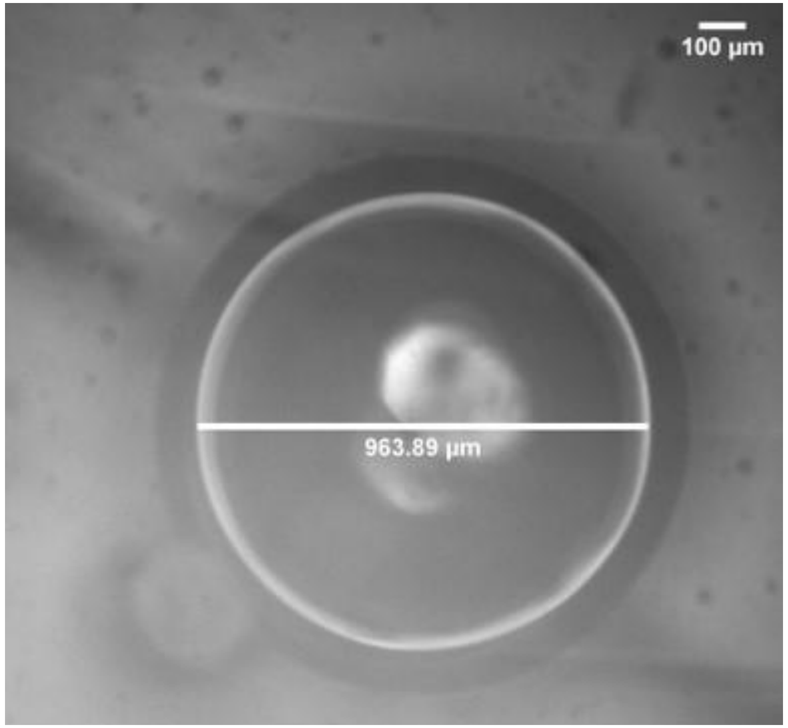
Continuous phase rate: $250\mu L/min$

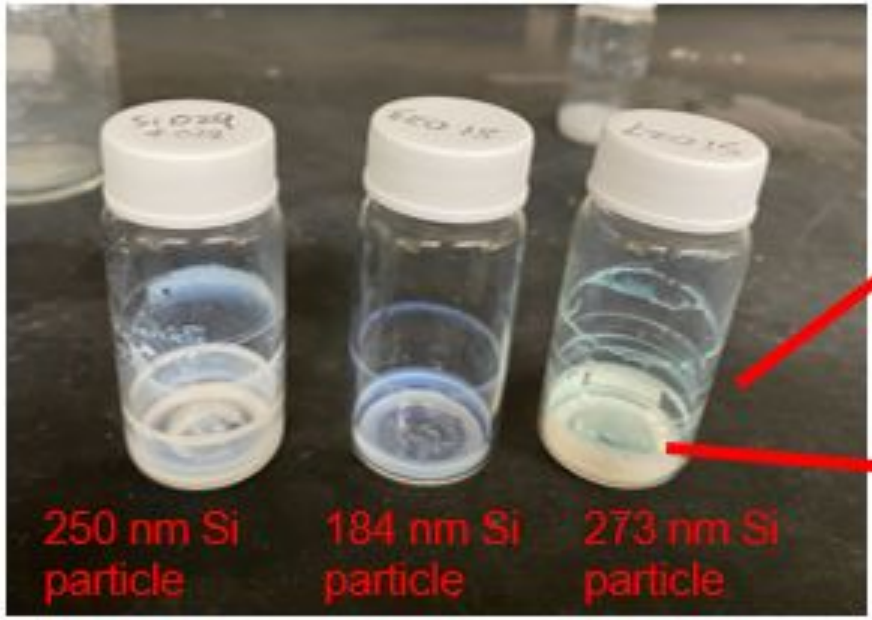
Problem: Air bubbles leaking in channels, collapsing droplet

Future Solutions:

- Remake mold with different size
- channels for middle and continuous phase, and epoxy needles in place.

Double Emulsion, Observed Color

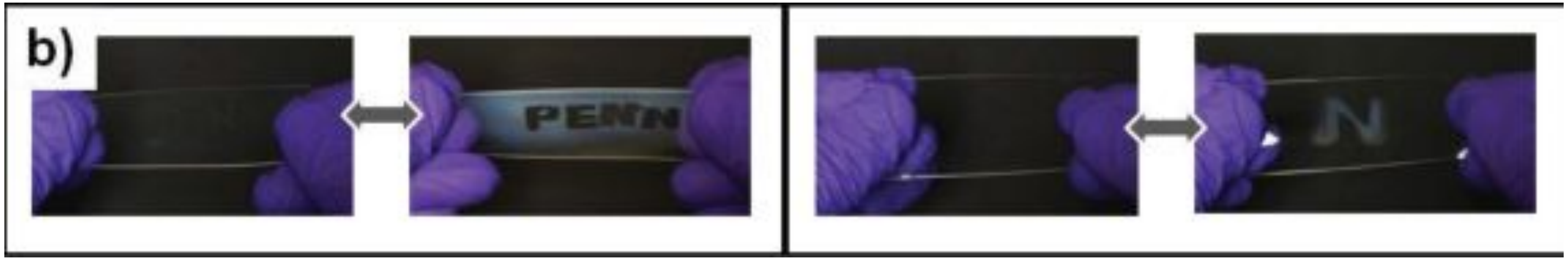




Applications

The actual application and significance of studying this is seeing how these nanoparticles or structures work to make structural color. The idea of these nanostructures can be very

- useful for photonic application. Such as
- stimulus-responsive sensors
  - Humidity sensors
- reflective displays
- light harvesting of solar cells
- smart windows.
  - become opaque to block or reflect sunlight , then return to transparent at low lighting conditions

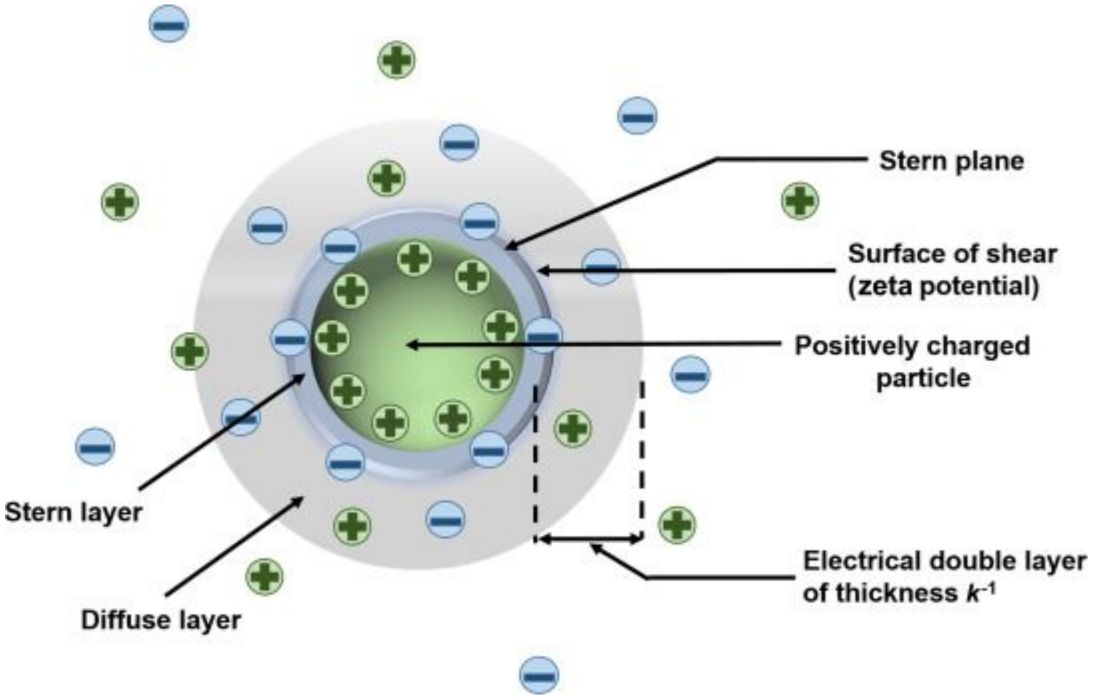


Ge D, Lee E, Yang L, Cho Y, Li M, Gianola DS, Yang S. A robust smart window: reversibly switching from high transparency to angle-independent structural color display. Adv Mater. 2015 Apr 17;27(15):2489-95. doi: 10.1002/adma.201500281. Epub 2015 Mar 2. PMID: 25732127.

Conclusions

Reasons as to why it didn't work?

- Volume Fraction too high
- Aggregation of particles
- No hydrophobic ends
- Pre-Order Molds Microfluidics
- Try different cocktails!
- Find a way for there to be a repulsive force on particles



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